

Brac University
Department of Electrical & Electronic Engineering
Spring2026

Course Number : EEE308L

Section :06

Group No :04



PSpice Lab Report 2

Name: Souvik Barman Ratul

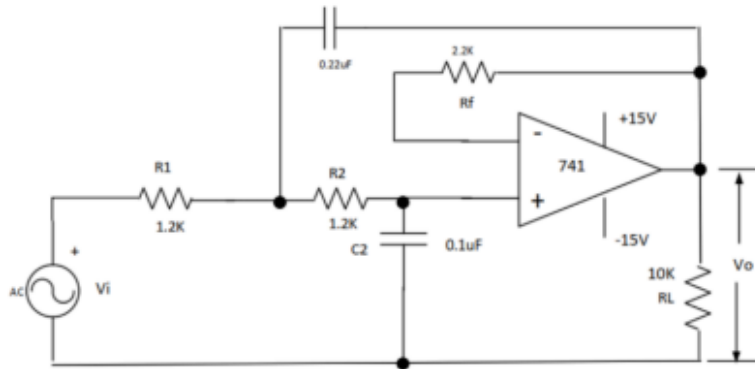
ID:24121205

Section: 06

Semester:-Spring 2026

Problem-1

- 40 dB/Decade Low Pass Butterworth Filter

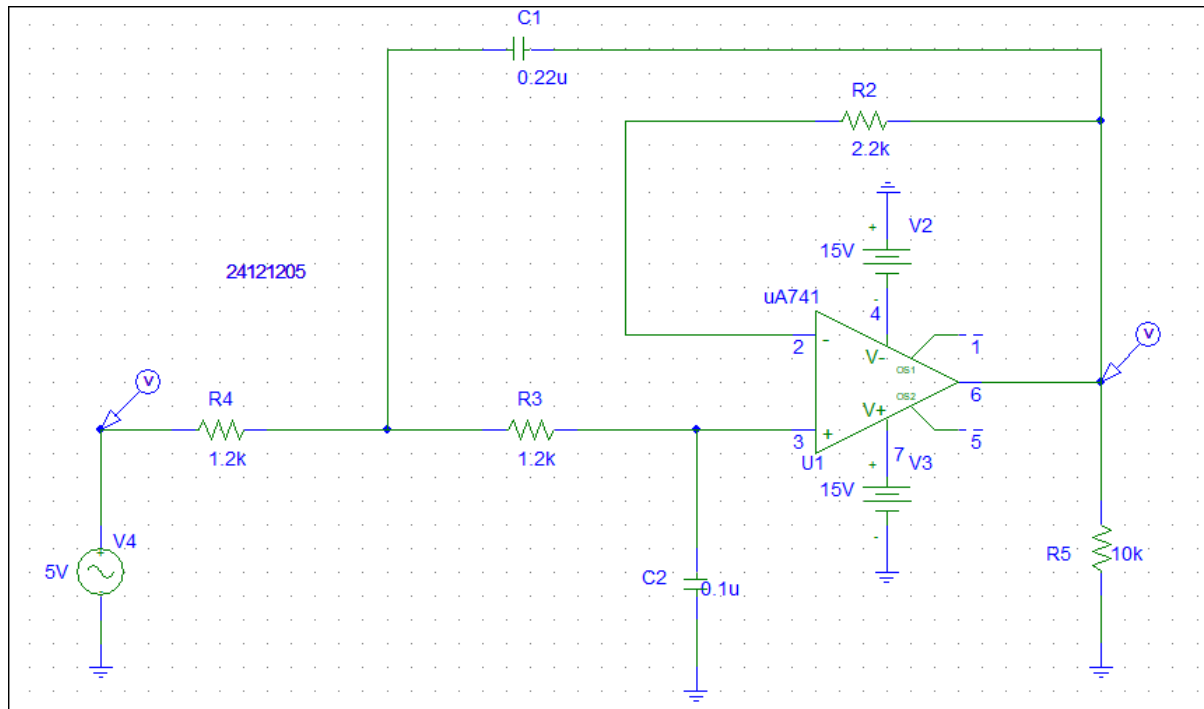


Draw the circuit in and give the source 5V amplitude and 0 phase.

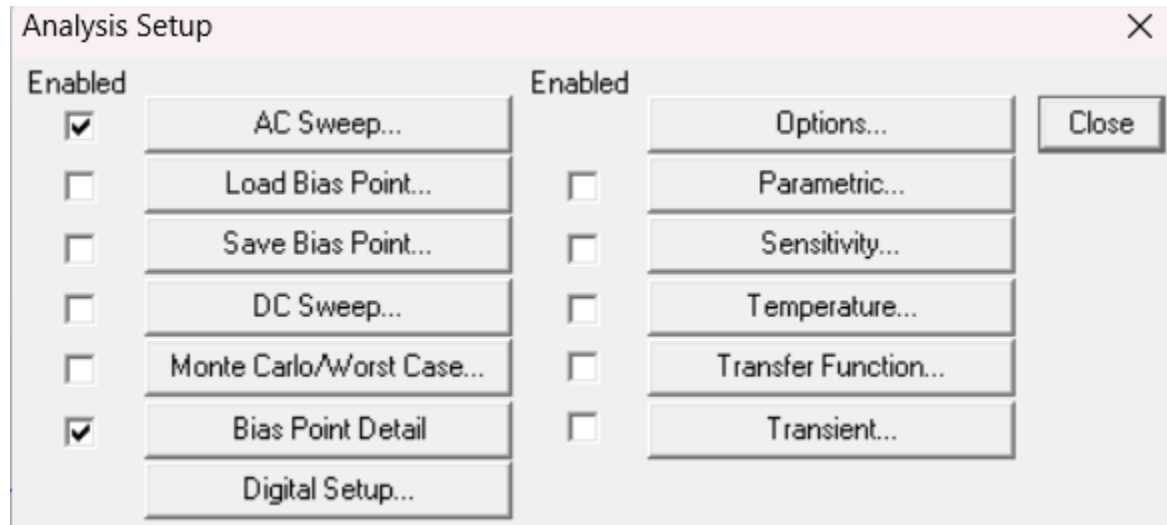
- (i) Plot the Gain and Gain(dB) with respect to frequency. Use a suitable frequency range.
- (ii) From both the Gain and Gain(dB) plot, find the cutoff frequency.
- (iii) Show that the roll-off is -40dB/Decade.

Answer:-

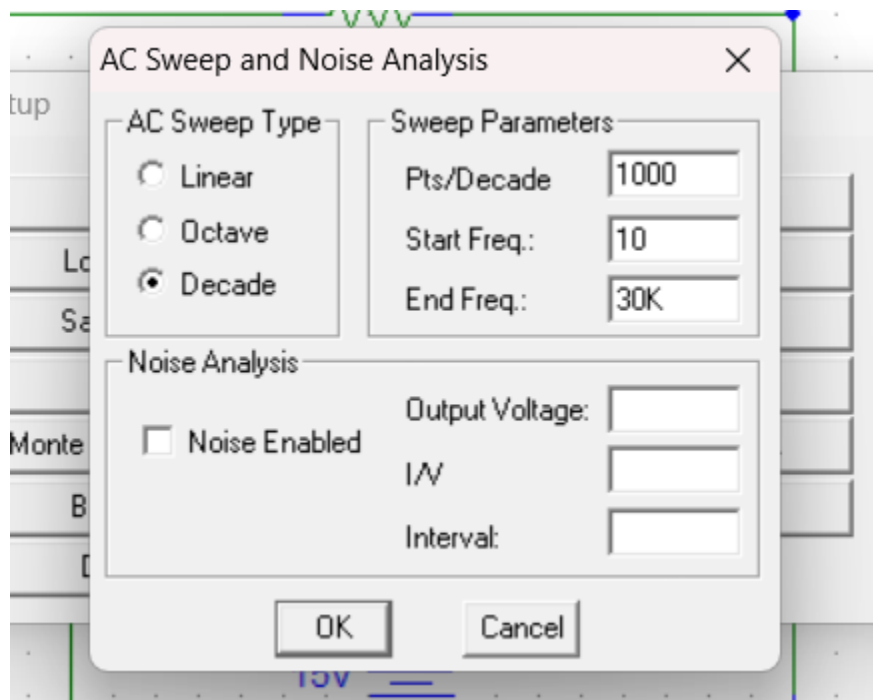
(i) **Circuit Diagram-**



Analysis Setup:



AC Sweep Setup



VAC parameter

V3 PartName: VAC ✕

Name	Value
ACMAG	= 5V

DC=0V
* SIMULATIONONLY=
* PART=VAC
ACMAG=5V
ACPHASE=
* MODEL=
PKGREF=V3

Save Attr

Change Display

Delete

OK

Cancel

☒ Include Non-changeable Attributes

☒ Include System-defined Attributes

Graph

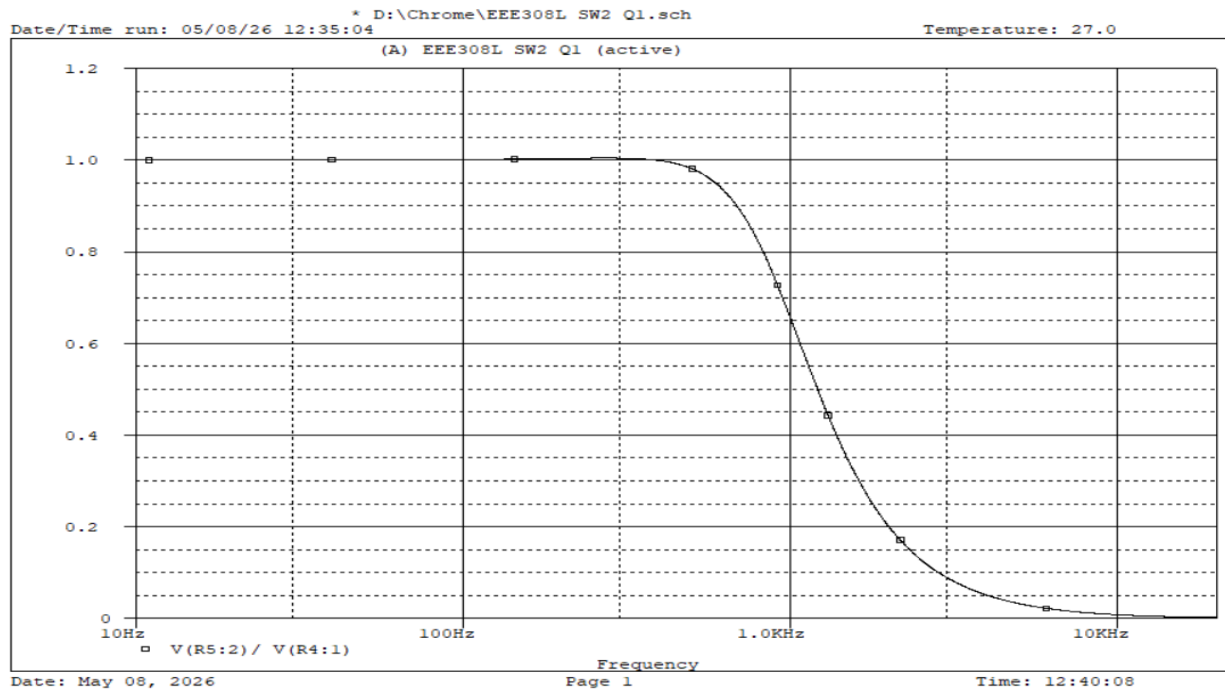


Figure: Graph for Gain

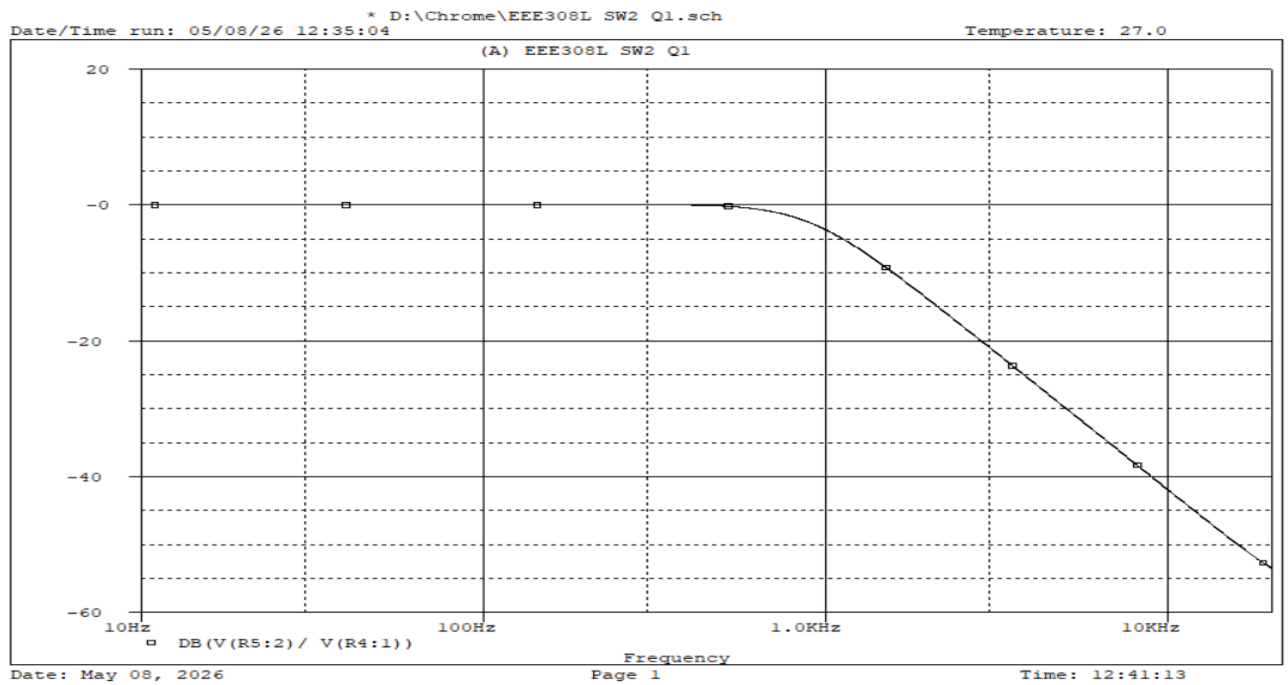


Figure: Graph for Gain (dB)

(ii)

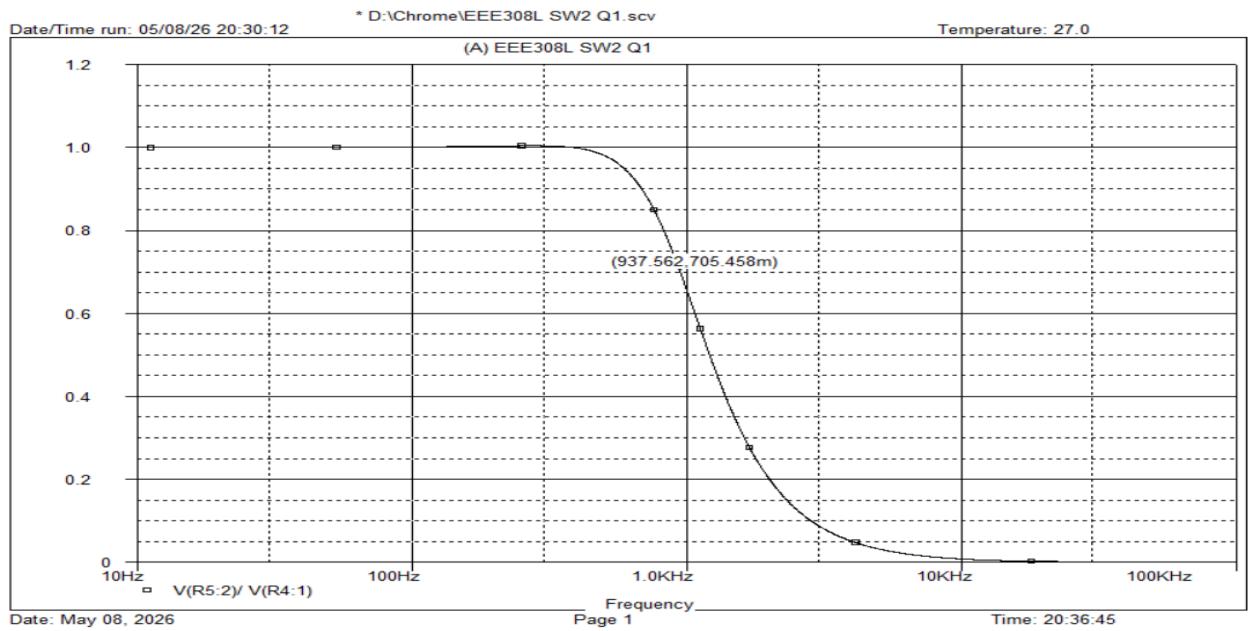


Figure: Graph for Gain (cutoff $f = 937.562\text{Hz}$)

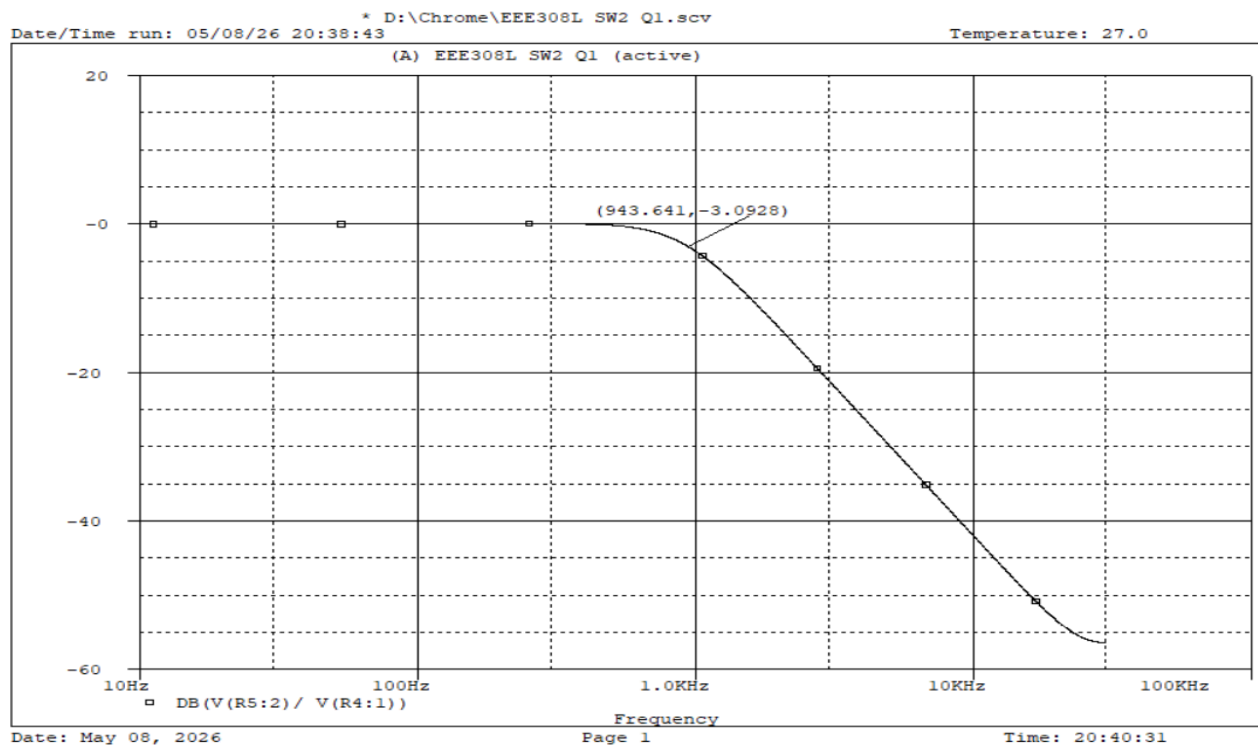
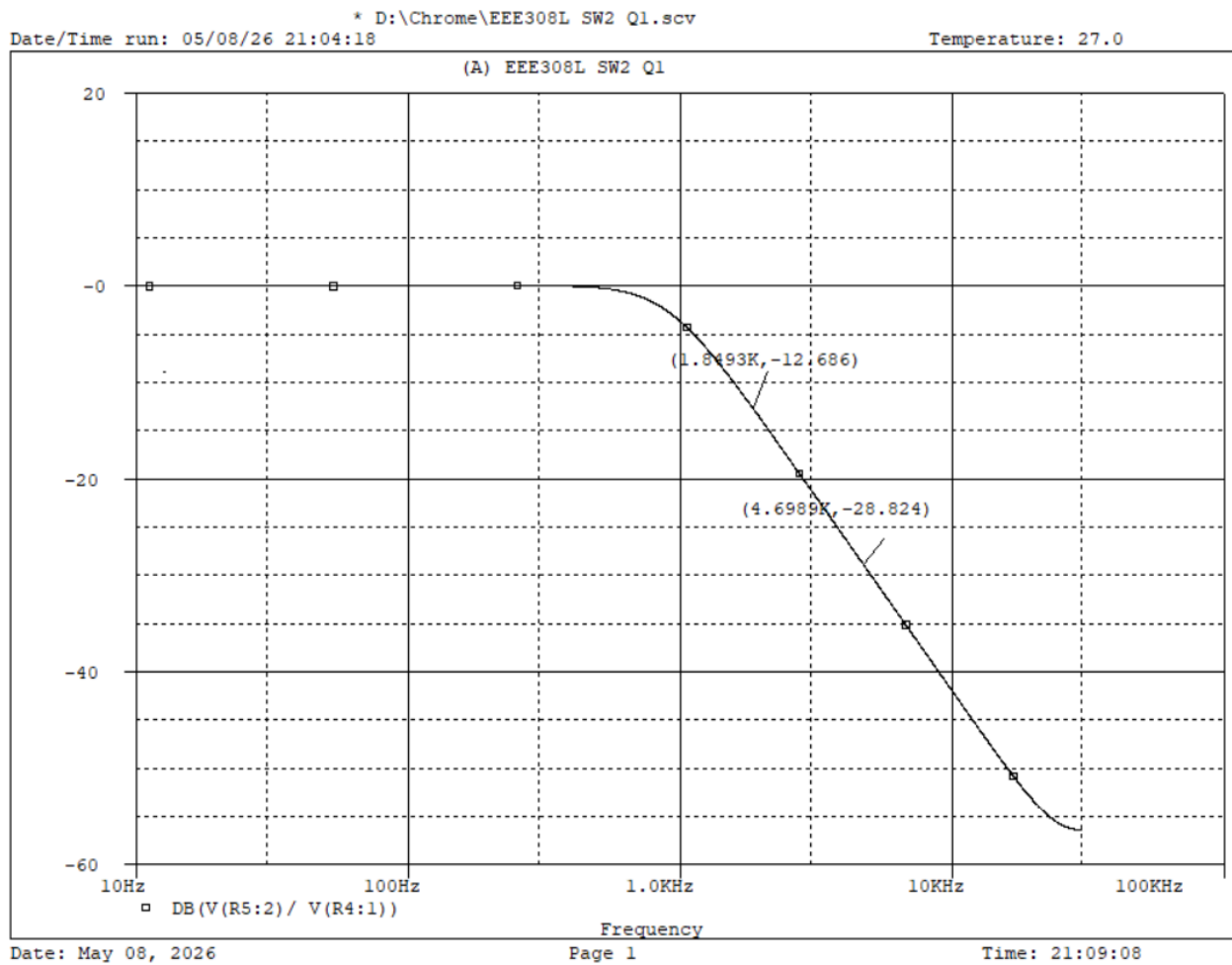


Figure: Graph for Gain (dB) $f = 943.641\text{Hz}$ at $\text{dB} = -3.09$

(iii)

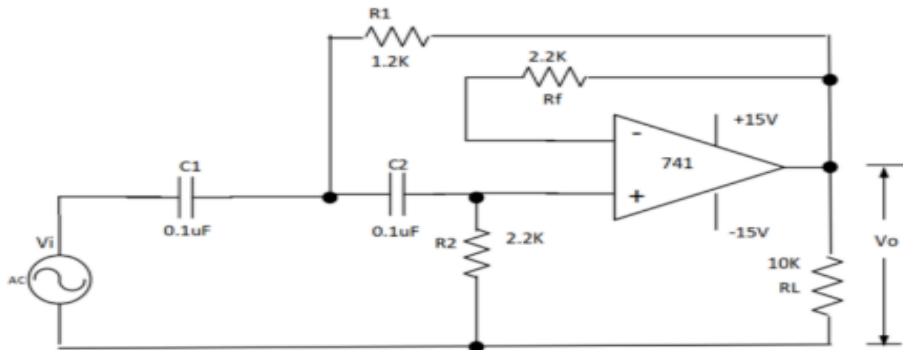


Using 2 different points in the function:

$$\text{Roll off} = \frac{-28.824 - (-12.686)}{\log(4.6989) - \log(1.8439)}$$
$$= -39.72 \approx -40\text{dB/Decade (Ans).}$$

Problem-2

40 dB/Decade High Pass Butterworth Filter

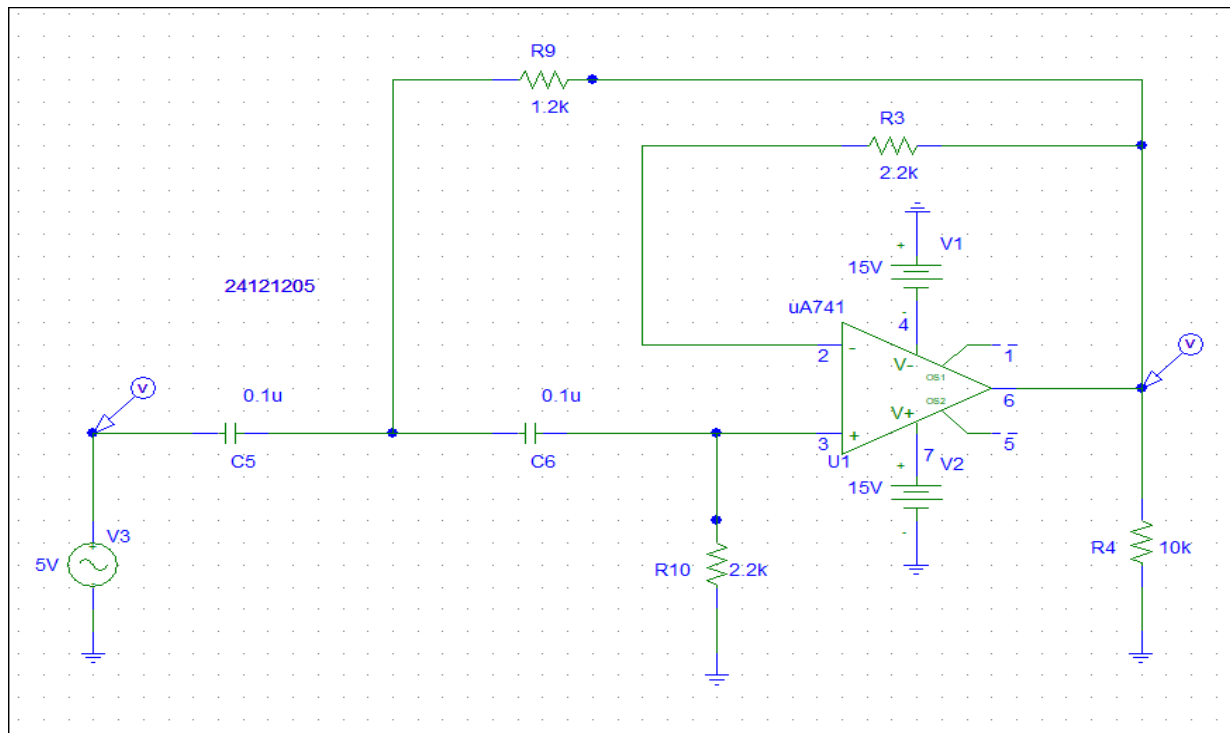


Draw the circuit in and give the source 5V amplitude and 0 phase.

- Plot the Gain and Gain(dB) with respect to frequency. Use suitable frequency range.
- From both the Gain and Gain(dB) plot, find the cutoff frequency.
- Show that the roll-off is 40dB/Decade.

Answer:-

(i) **Circuit Diagram-**



VAC parameter

V3 PartName: VAC

Name	Value
ACMAG	= 5V

DC=0V
* SIMULATIONONLY=
* PART=VAC
ACMAG=5V
ACPHASE=
* MODEL=
PKGREF=V3

☒ Include Non-changeable Attributes
☒ Include System-defined Attributes

Save Attr
Change Display
Delete
OK
Cancel

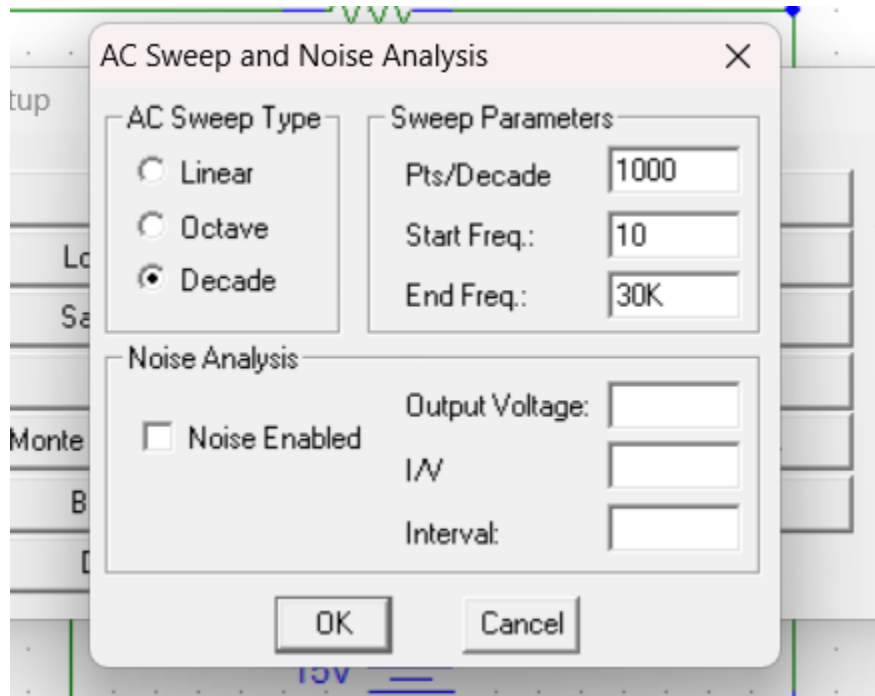
Analysis Setup:

Analysis Setup

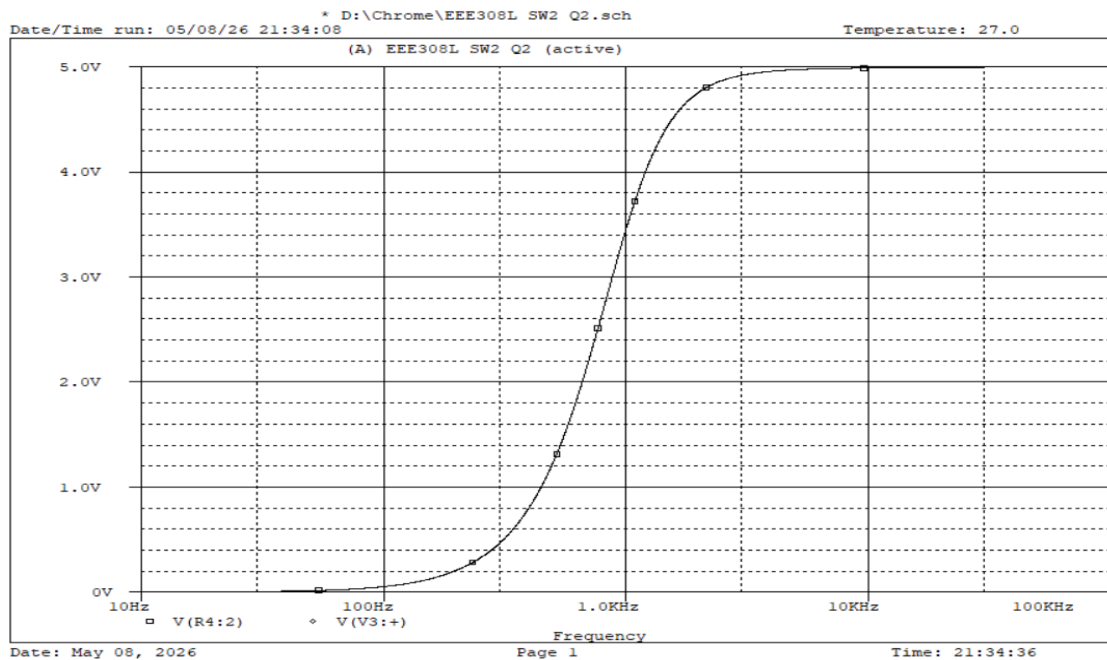
Enabled		Enabled	
☒	AC Sweep...	☐	Options...
☐	Load Bias Point...	☐	Parametric...
☐	Save Bias Point...	☐	Sensitivity...
☐	DC Sweep...	☐	Temperature...
☐	Monte Carlo/Worst Case...	☐	Transfer Function...
☒	Bias Point Detail	☐	Transient...
	Digital Setup...		

Close

AC Sweep Setup



Graph



(ii)

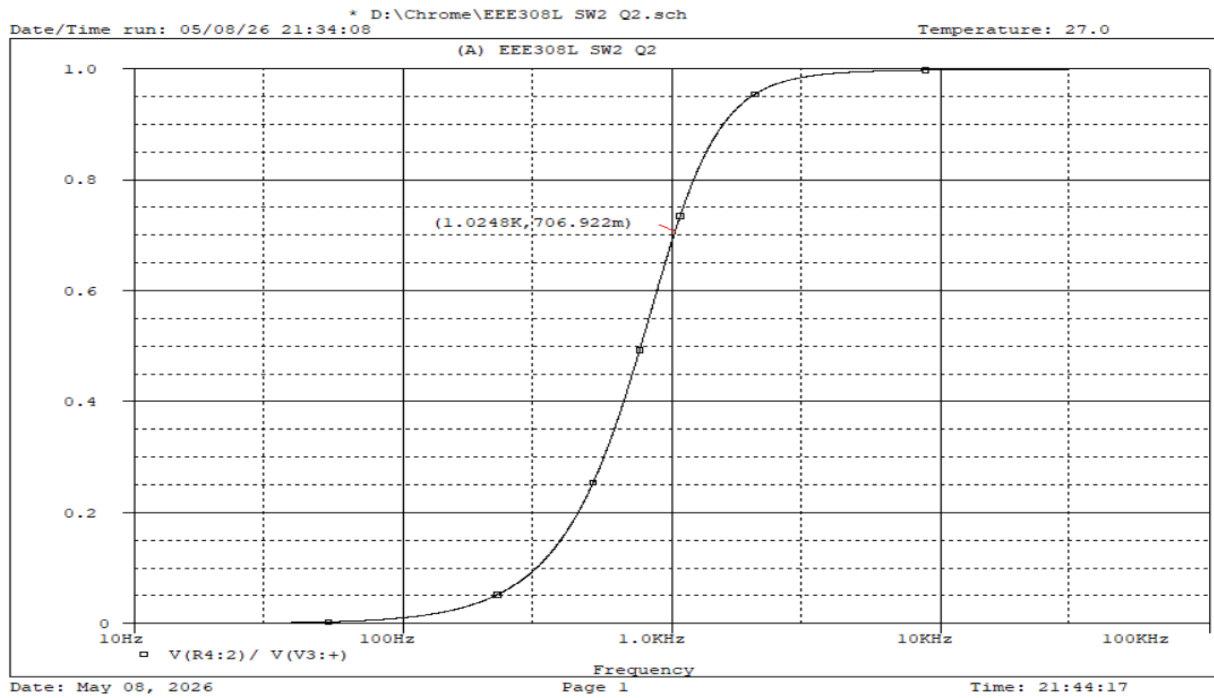


Figure: Graph for Gain (cutoff $f = 1.245\text{Hz}$) at $0.707v$ Gain

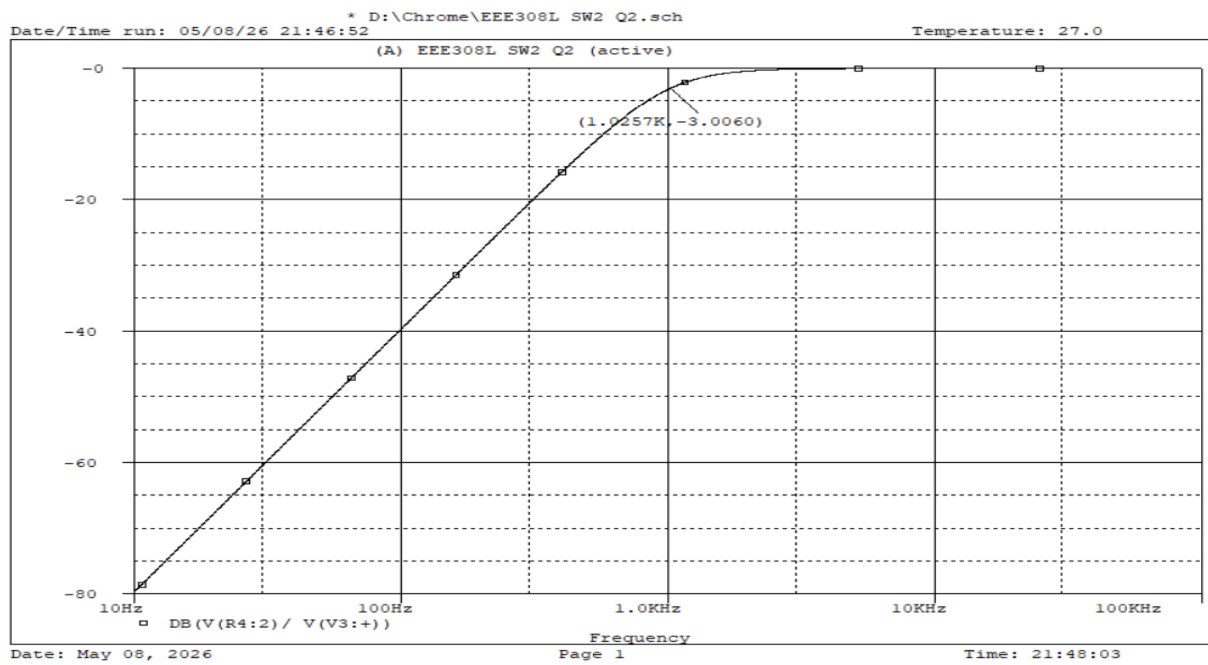
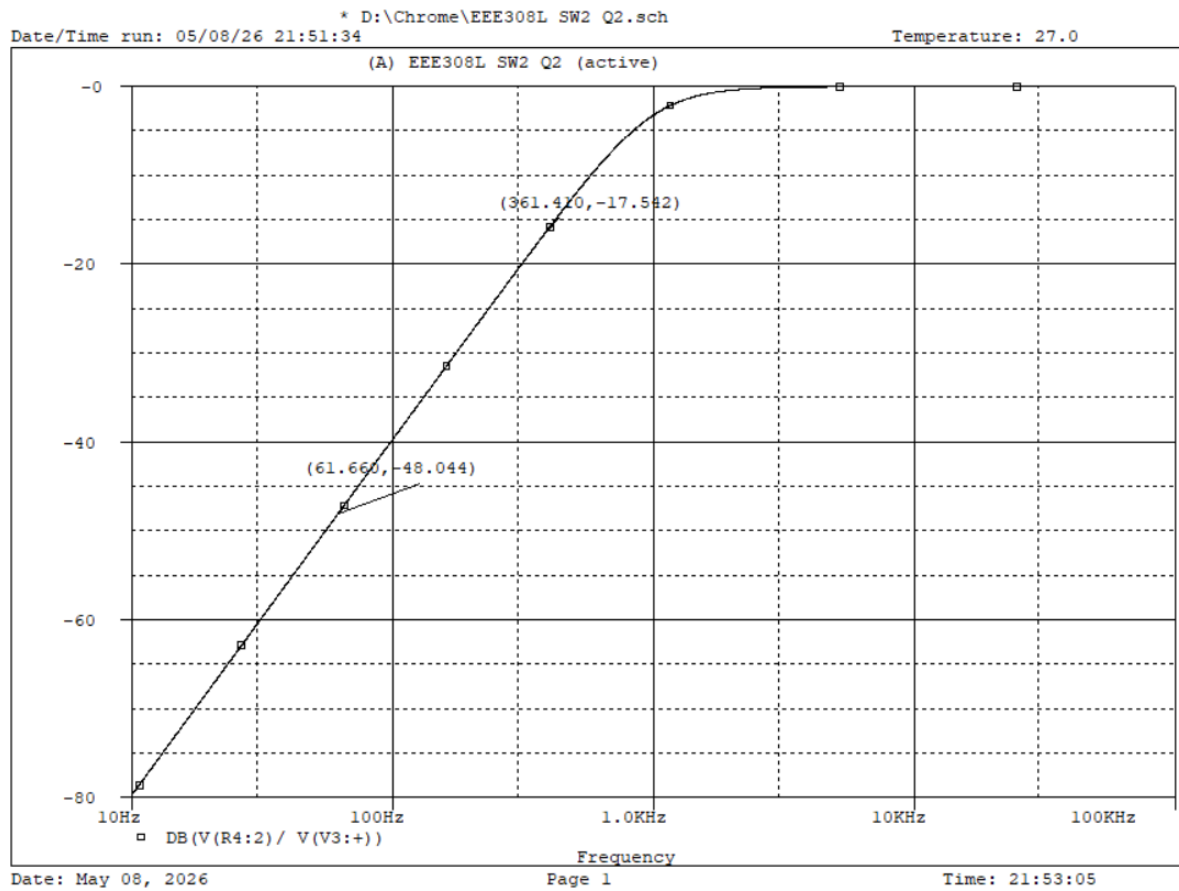


Figure: Graph for Gain (dB) $f = 1.0257\text{Hz}$ at $\text{dB} = -3.006$

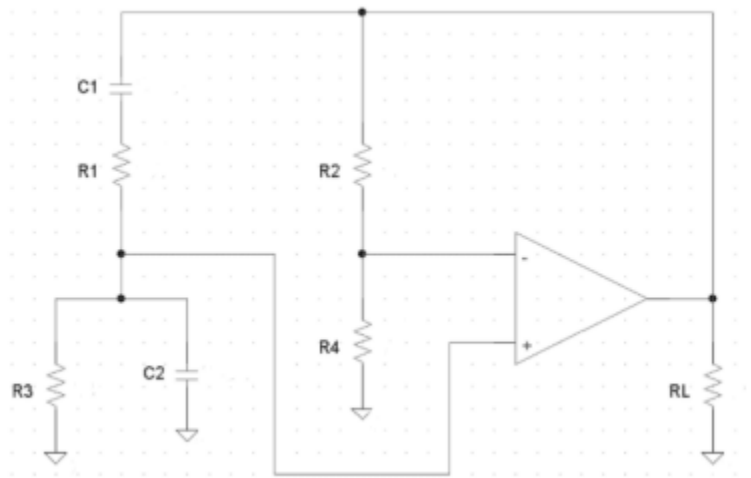
(iii)



Using 2 different points in the function:

$$\begin{aligned}\text{Roll off} &= \frac{-48.044 - (-17.542)}{\log(361.410) - \log(61.660)} \\ &= -39.71 \approx -40\text{dB/Decade (Ans).}\end{aligned}$$

Problem-3



Design the stable sine-wave Wien Bridge oscillator for an audio-testing application. The target frequency is $20x$ where x = last 2 digits of your ID.

Answer:-

Student Id= 24121205

Here , $f = 2005$

For Wien Bridge Oscillator, Frequency, $f = 1/2\pi RC$

Set, $c = 0.22\mu F$

$R = 1/2\pi fC$

$$= 1/2\pi * 2005 * 0.22 * 10^{-6}$$

$$= 3608.137 \Omega$$

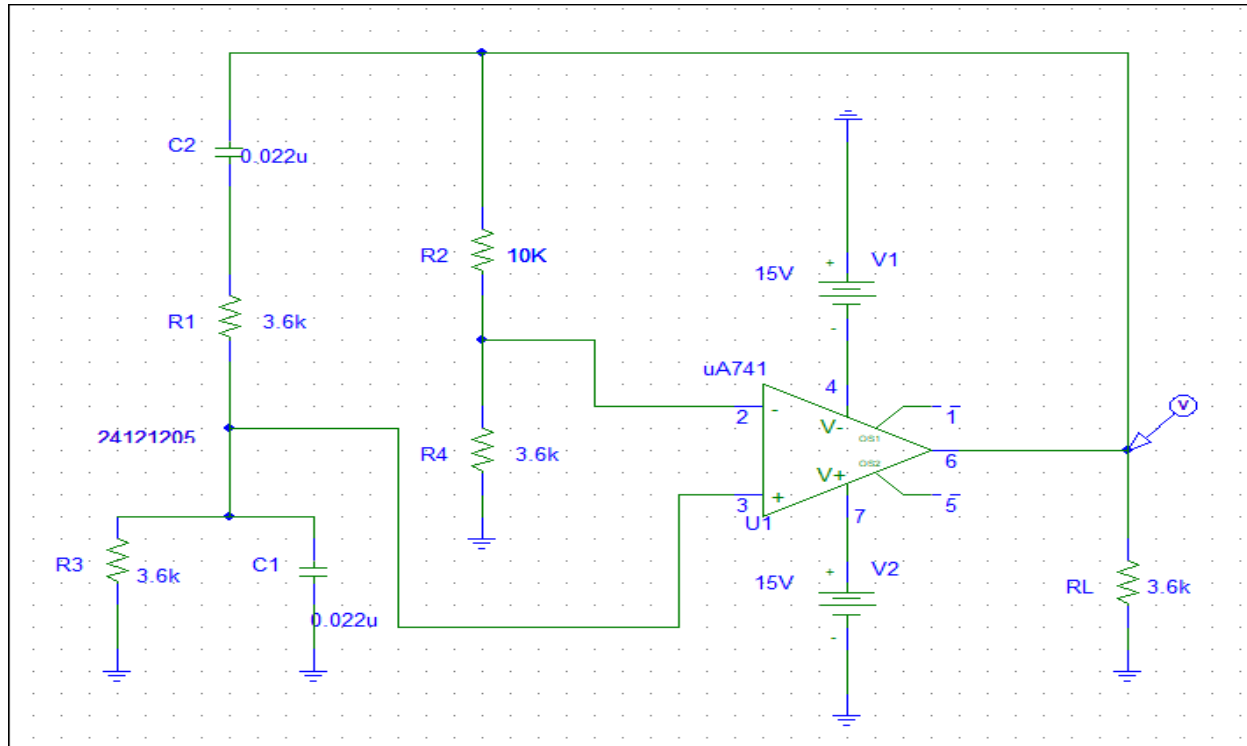
$$= 3.6k\Omega$$

To get a stable oscillation,

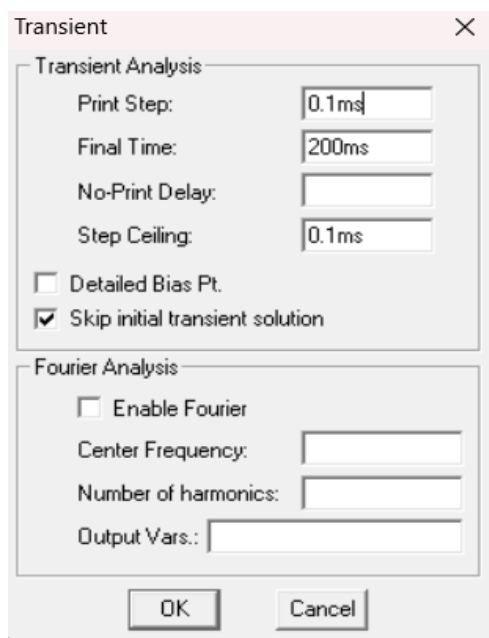
we need a gain over 3, So, we set $R2 = 10k\Omega$

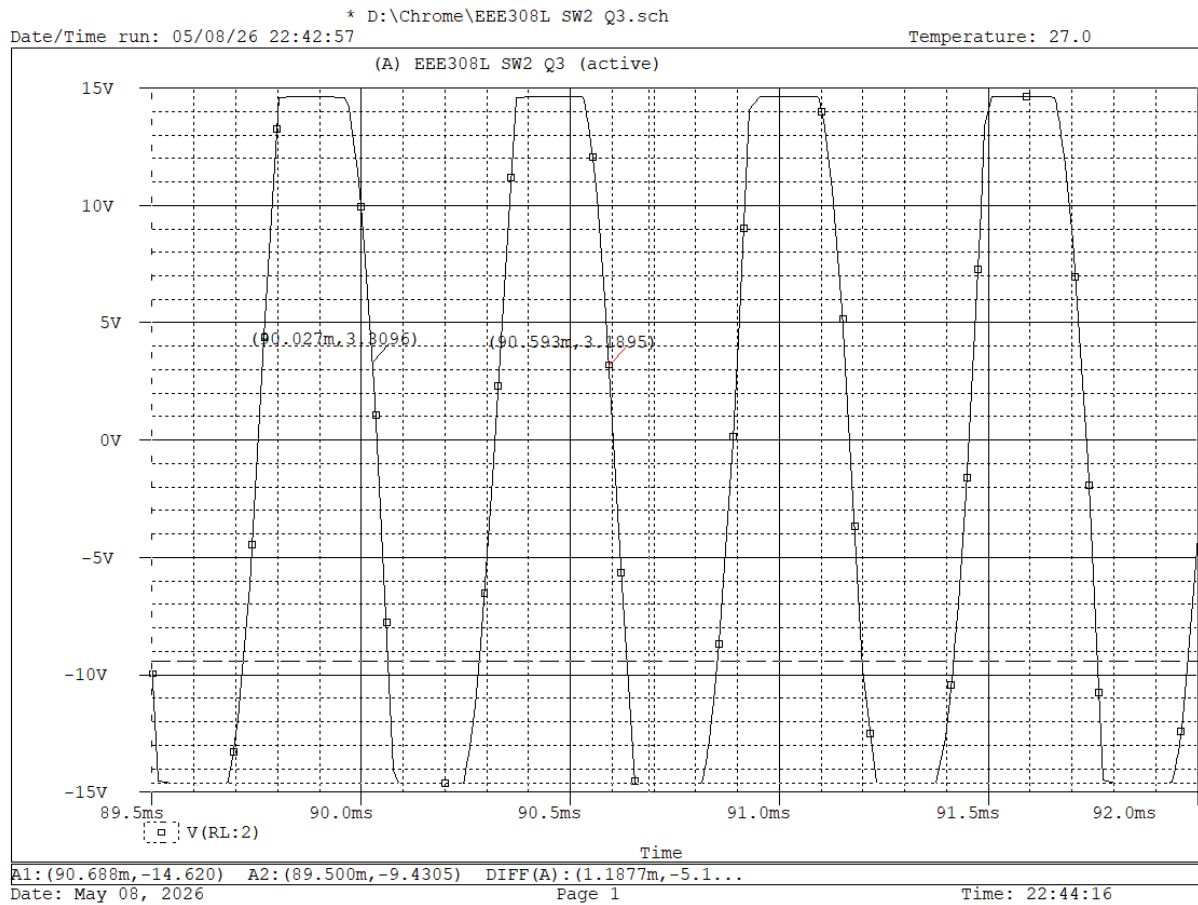
$$\text{Gain, } A_v = 1 + (R2/R1) = 1 + (10/3.6) = 6.5$$

Circuit diagram



Transient Setup





Here,

$$T = (90.593 - 90.027) \text{ms}$$

$$\text{So, } T = 0.566 \text{ms} = 566 \text{ us}$$

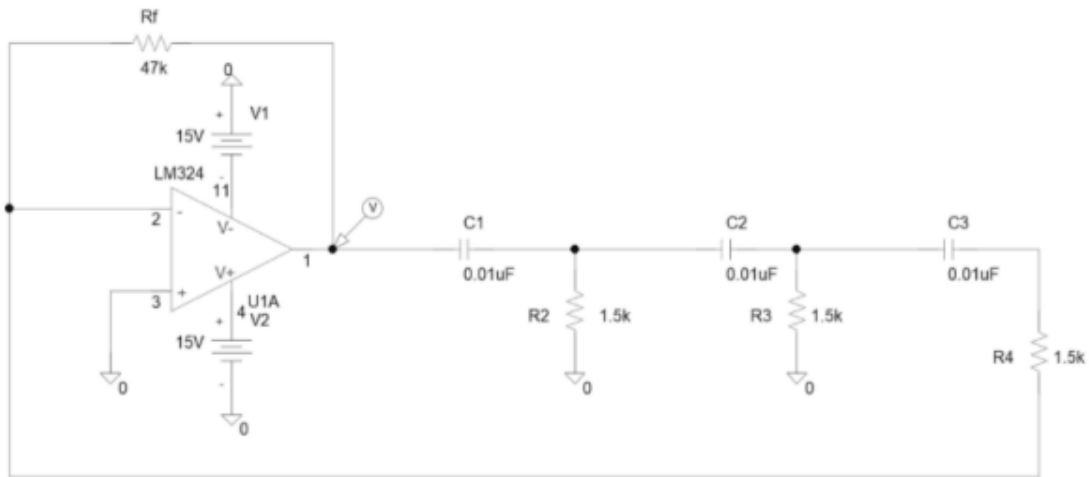
$$f = 1/T = 1/566 \times 10^{-6}$$

$$= 1766.8 \text{Hz}$$

$$\text{Error} = ((2005 - 1766.8)/2005) \times 100\%$$

$$= 11\%$$

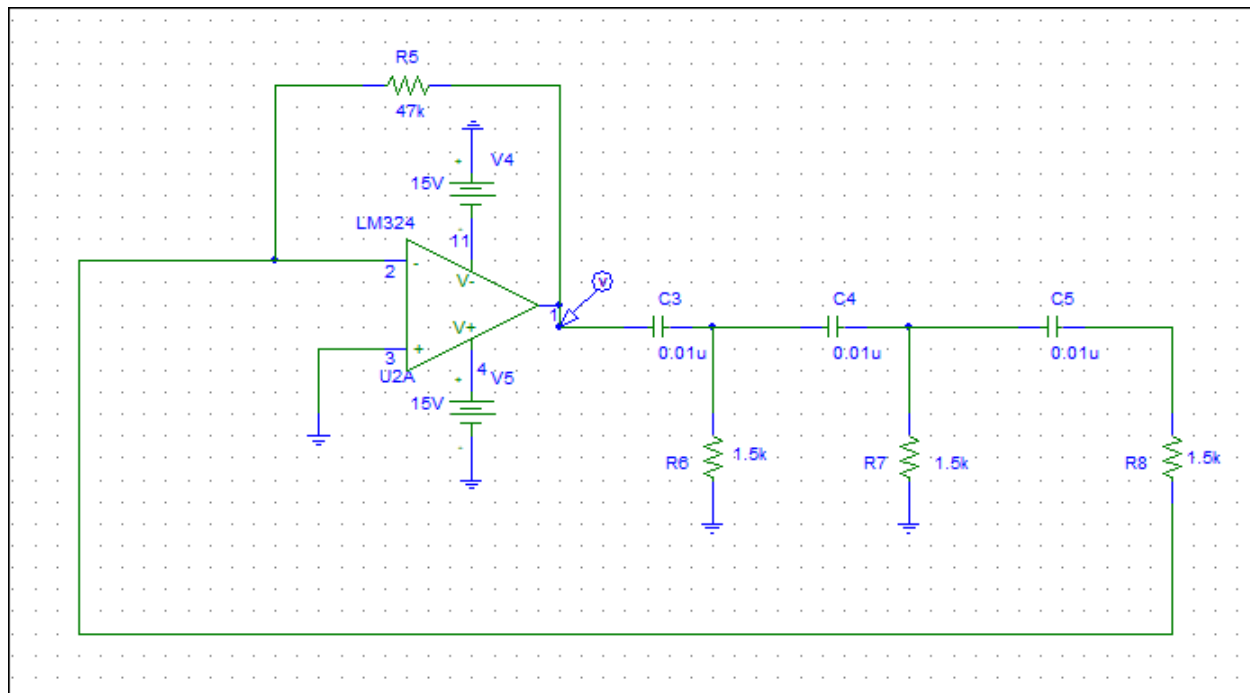
Problem-4



- (i) Construct the above circuit.
- (ii) Run the simulation in the 'Bias Point Detail' and 'Transient' mode.
- (iii) Observe the wave shapes and determine the frequency.

Answer:-

Circuit Diagram:



Transient Setup

Transient

Transient Analysis

Print Step: 0.01ms

Final Time: 100ms

No-Print Delay:

Step Ceiling:

☐ Detailed Bias Pt.

☒ Skip initial transient solution

Fourier Analysis

☐ Enable Fourier

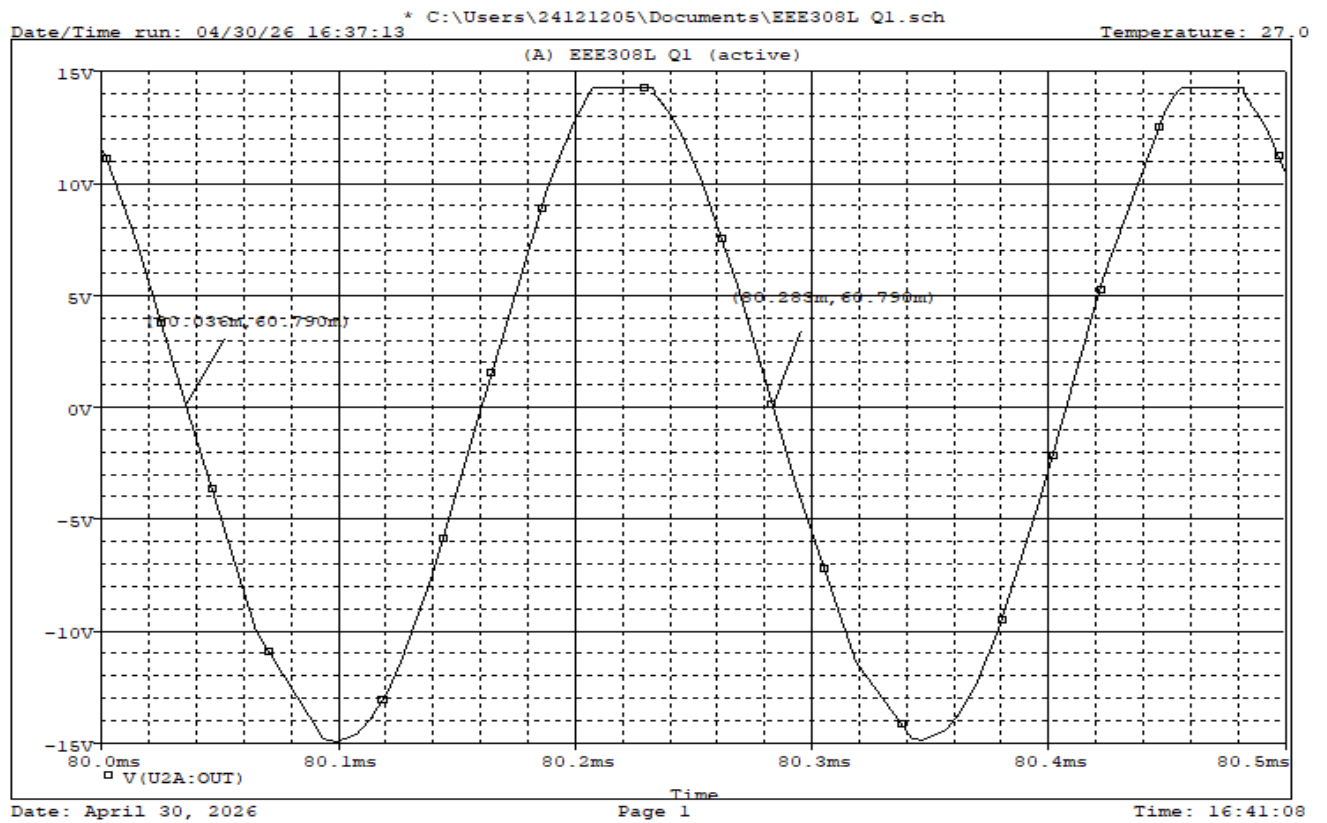
Center Frequency:

Number of harmonics:

Output Vars.:

OK Cancel

V_o Graph



Here,

we find time period between 1 full period,

$$T = (80.283 - 80.036) \text{ms}$$

$$= 0.247 \text{ms}$$

$$= 247 \mu\text{s}$$

We know,

$$\text{frequency, } f = \frac{1}{T}$$

$$f = \frac{1}{247 \times 10^{-6}} = 4048.582 \text{Hz} (\text{Ans})$$